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Analysis of critical raw materials policy for electrical and electronic equipment: Planning for a truly circular economy

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A B S T R A C T

The European Union's objectives under the Green and Digital transitions, are contingent on the decarbonisation of essential sectors such as energy, transportation, and communications. At present, the EU does not have sufficient identified mineral resources, nor processing capacity, to supply many of its strategically defined materials. The success of these objectives hinges on the EU's ability to secure a stable and consistent supply of critical raw materials (CRM). Using a systematic three-part assessment of mandated and non-mandated EU policies, European Commission communications, and a review of CRM, the findings reveal that despite efforts to enhance local supply and diversify foreign sources of CRM, there is a lack of cross-sector coherence across social, economic, and technical aspects of recycling. This is further exacerbated by inefficient retention of waste-hosted critical raw materials. Failure to adopt integrative policy measures may result in the EU's continued dependency on undiversified foreign sources for its supply of critical raw materials, thereby compromising its strategic autonomy as well as the goals of the Green and Digital transitions. Improved design of electrical and electronic equipment and processing of waste from these products could conceivably provide a mechanism for a continuous and retained supply of critical raw materials, culminating in a potentially significant resource stream for the Union. The EU can improve resource efficiency, reduce its dependence on imports, and facilitate a truly circular economy by adopting a comprehensive and symbiotic policy framework that quantifies material resources in waste and that recognises the product design stage as an integral facet of the product recycling stage, herein referred to as "forwardcycling".

1. Introduction

Critical raw materials (CRM) such as cobalt (Co), lithium (Li) and the rare earth elements (REE) are essential components in renewable energy technologies and high-tech applications. The European Union (EU) defines CRM as substances that "*reach or exceed thresholds for both economic importance and supply risk*" (European Commission, 2018). In the EU, these CRM are produced in limited quantities or subject to supply risks from both *primary*[1] and *secondary*[2] resources, which poses a significant challenge to both the EU's circular economy targets and the United Nation's Sustainability Goals (European Commission, 2020a; Muench et al., 2022). As the world transitions to a greener future, the demand for these materials is set to place strain on an already limited global supply (Ragnarsdóttir, 2008; European Commission, 2020a). For example, the demand for European Li, is projected to grow about 12 times by 2030 and around 21 times by 2050. Similarly, the demand for neodymium (Nd) and dysprosium (Dy), both REE, is expected to increase approximately 5–6 times by 2030 and up to 6 to 7 times by 2050 (Carrara et al., 2023), raising concerns about the viability of the Green and Digital transitions as well as potentially adverse effects on EU competitiveness, employment, working conditions, and wages (Official Journal of the European Union, 2022; European Commission, 2023a).

What is deemed a critical raw material is subject to change through time and lists vary from country to country depending on each country's assessment of economic importance and supply risk. Currently in the EU, 34 metals, metal groups, and non-metallic minerals are listed as CRM (Fig. 1; European Chemical Society, 2020). Those recognised as important to areas of national and international strategy (i.e. essential to their technologies supporting the energy and digital transitions and the EU's defence and aerospace objectives) are classified as Strategic Raw Materials (SRM; Fig. 1; European Commission, 2023b). Many of the EU's CRM and SRM are sourced from primary resources, such as rocks/minerals (e.g. bauxite, feldspar, fluorspar, etc.) and/or metallic elements and their ores, with the mining and processing of most of these materials highly concentrated outside of the EU (European Commission, 2023c). Although there are significant issues surrounding primary

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[1] Primary resources are finite and non-renewable, naturally occurring materials that are extracted from the earth's crust through mining or other processes and are then processed to obtain the desired minerals or metals.

[2] Secondary materials/metals/resources are those produced or extracted from the breakdown and processing of waste/discarded products, including, but not limited to, electronic waste.

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resource supply, this study focuses on EU policy surrounding secondary resources, or those occurring in waste streams.

Whilst CRM occur in various waste streams, this study focuses on waste electrical and electronic equipment (WEEE). WEEE materials largely comprise discarded electronic devices ranging from electronic vaporisers (i.e. e-cigarettes) to laptops to wind turbine generators (European Parliament, 2023a), all of which hold increased potential for providing a major source of multiple CRM. In fact, approximately 68% of the raw materials required to manufacture smartphones (Fig. 1), tablets, and laptops are classified as critical, whilst around 44% are considered strategic (Carrara et al., 2023). Adequate waste policies could encourage development and utilisation of innovative recycling technologies that could augment the CRM supply chain for continued production of high-tech applications (Kleijn and Van der Voet, 2010; Hagelüken et al., 2012; Bloodworth, 2014). Current difficulties at accomplishing economic recycling hinge on the complexity of separation of valuable elements from WEEE. Given that the production phase of battery-powered devices bears the biggest carbon footprint (Fleck, 2024), policies that encourage manufacturing design that specifically and purposely enable more efficient end-of-life recycling are crucial. Integrating policies related to raw materials and WEEE recycling for CRM could also play an integral role in the EU's circular economy transition (Hartley et al., 2020) and may be key to retaining CRM within the EU.

Several studies have evaluated various aspects of CRM and WEEE. Hool et al. (2023) provide a thorough analysis of the European Critical Raw Materials Act (CRMA), detailing the policy's challenges and opportunities, and Cucchiella et al. (2015) discuss global trends in WEEE generation and recovery, highlighting the growing need for improved recycling technologies and policies. In the Global E-Waste Monitor, Baldé et al. (2017) provide valuable data on e-waste management practices and identify significant gaps in recycling efforts. These studies predominantly focus on recycling efficiency and technological advancements, with less attention given to the quantification of CRM as a resource within WEEE, and the effectiveness of CRM tracking and transparent manufacturing policies. This study addresses these gaps by investigating the unknowns around CRM content in WEEE and reviewing and evaluating the effectiveness of related policies, with a particular focus on WEEE manufacturing, processing, and recycling as a potential source of these important materials, as well as external influences on their implementation. The findings are particularly relevant to policy makers and mineral advocacy groups to better integrate and facilitate resource policies. It is important to note that this study focuses on the EU's CRM supply and its demand-driven socio-economic responsibilities and, as such, does not comprehensively consider potentially symbiotic policies and strategies of countries outside of Europe.

2. Methodology

A systematic three-part, semi-hierarchical structure was

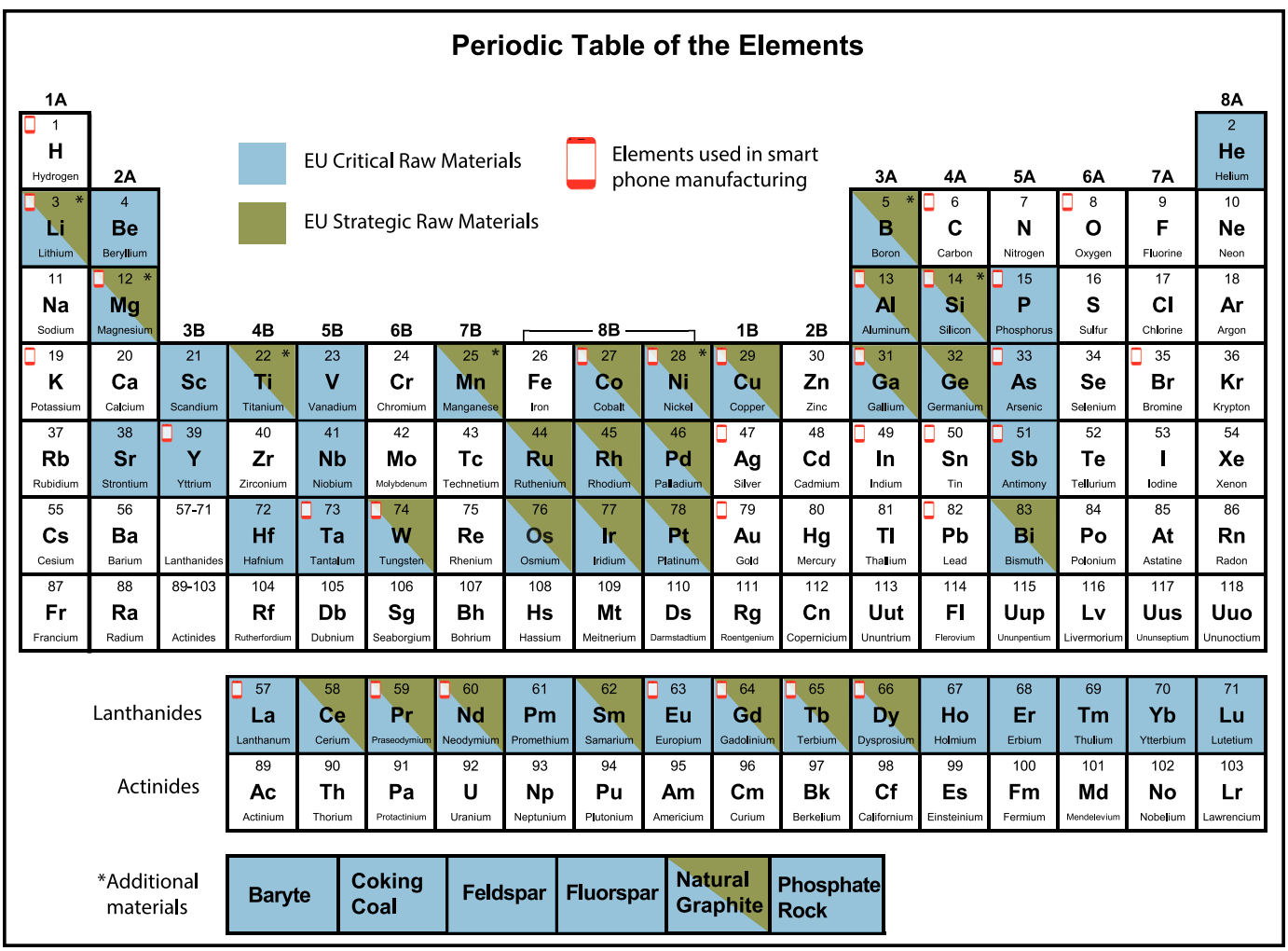

Fig. 1. Image showing Critical Raw Materials (blue) and Strategic Raw Materials (green) on the Periodic Table of Elements as well as additional, non-elemental materials, as defined by European Commission (2023b). * Boron: Metallurgical grade; Mg: Metal grade; Mn, Ni, Li: Battery grade; Si, Ti: Metal. CRM contained within mobile phones are indicated (European Chemical Society, 2020).

implemented to analyse existing EU policies related to CRM and waste processing. This was applied using data triangulation from three primary source categories, including mandated and non-mandated EU policies, European Commission communications, and academic literature. To further validate the results, the findings from the policy assessment and literature review are cross-referenced with reported data, allowing for a robust analysis of trends and gaps in existing and related CRM policies.

1. *Assessment of Mandated and Non-Mandated EU Policies*
   The study begins by reviewing mandated and non-mandated EU policies. It critically evaluates their effectiveness in securing "waste" CRM and optimising waste processing. The focus is placed on identifying opportunities for improvement within the EU framework.
2. *Analysis of European Commission Communications*
   To understand the evolution of CRM policies and their alignment with broader EU directives, this study systematically analyses European Commission communications related to CRM policy. Starting from the early 2000s, with the introduction of key policies such as the WEEE Directive 2002/96/EC and the RoHS Directive 2002/95/EC (both of which have since been revised), to present day, this analysis examines correlations between past, current, and future policy initiatives and existing directives.
3. *Literature Review*
   In parallel, the study conducts a review of relevant research and academic literature pertaining to CRM and the external factors that may affect the supply chain. This step aims to identify strengths, weaknesses, and gaps in current EU policy.

3. WEEE as a resource

Global reports indicate that WEEE grows at an annual rate of approximately 3–5% (Cucchiella et al., 2015). The EU experienced a notably higher annual WEEE growth rate of roughly 6.8% between 2012 and 2021. During this period, the total amount of electronic equipment put on the EU market increased by approximately 77%, from 7.6 million tonnes in 2012 to 13.5 million tonnes in 2021. Moreover, the total collected WEEE in the EU also saw significant growth, rising from 3 million tonnes in 2012 to 4.9 million tonnes in 2021, reflecting a 65% increase, with a compound annual growth rate (CAGR; Brealey et al., 2020) of 5.2% over this period for collected WEEE (Eurostat, 2023; Eurostat, 2022). This amounts to approximately 64% of all electronic equipment during that period unaccounted for or not collected.

However, despite its increasing accumulation rate, technical challenges often impede the effective utilisation of WEEE as a resource. Recycling rates vary among different materials, with elements like aluminium (Al), iron (Fe), lead (Pb), and zinc (Zn) being recycled at rates exceeding 50%. Whilst others, like most REE and other CRM, face significant hurdles in recycling due to technical complexities and economic viability, barely reaching recycling rates of 1–3% (Fig. 2; Blengini et al., 2019; European Commission, 2020a; Geng et al., 2021; Li et al., 2022; Jowitt, 2023). WEEE presents a significant challenge to recycling as it contains a multitude of elements, making material heterogeneity a prominent issue. Recycling of CRM such as boron (B), Co, gallium (Ga), germanium (Ge), hafnium (Hf), Li, niobium (Nb), scandium (Sc), strontium (Sr), tantalum (Ta), and the REE, specifically Nd, from WEEE and other waste, whilst technically feasible, is not currently undertaken at significant scale (Blengini et al., 2019; Geng et al., 2021; Li et al., 2022; Jowitt, 2023). Material Systems Analysis reports (BIO by Deloitte, 2015; European Commission, 2023d) illustrate that while the collection rate for REE-bearing materials is quite high, particularly for REE-bearing permanent magnets, the actual end-of-life recycling input rates (EOL-RIR) and recycling rates (EOL-RR) are quite low (Fig. 2) and can even show variability across reports (e.g. compare Talens Peiró et al., 2018 and European Commission, 2023d). In some instances, REE may technically be recycled, but in reality they are "non-functionally recycled" as they are tied up in other recycled materials and not the intended target resource (BIO by Deloitte, 2015).

Many industry-reported instances of CRM recycling, particularly for REE, are better characterised as reusing or upcycling,[3] as seen with companies like Apple. Apple has developed several specialised robotics systems that disassemble specific components for reuse. They also have specialised shredders that break down old devices whilst still preserving REE-bearing permanent magnets for direct re-use in new devices (Apple, 2021).

Recycling, especially for REE and most of the other CRM, is a "wicked problem[4]" (Head, 2022) plagued with a number of interconnected issues including, but not limited to: the heterogeneity of materials; a lack of information on the elemental concentrations in past and present consumer items, landfills, and recycling centres; time delays in recycling streams; the economic viability of recovery processes, processing efficiency and material preference for recycled materials; and the variability in product EOL-RIR (see Fig. 2). WEEE items encompass a wide range of products which, in turn, comprise a variety of components containing a multiplicity of elements. For example, a smartphone contains at least trace amounts of over half of the elements in the Periodic Table (European Chemical Society, 2020, Fig. 1). The recycling process is further complicated by the lack of comprehensive data on elemental concentrations in consumer products, a result of insufficient policies and regulations mandating detailed reporting and tracking of these elements. Not surprisingly, there has been a focus on capturing elements, such as platinum (Pt) and gold (Au), which are technologically easier to separate and more profitable (Masterson, 2021). The capture of more challenging elements, such as REE from WEEE, has made little to no progress, in-part due to the absence of stringent regulations that could incentivise research and development in this area (Reck and Graedel, 2012; Hagelüken, 2014; Huisman et al., 2017; Lixandru et al., 2017; European Commission, 2020a; Geng et al., 2021).

4. EU policies and circular economy complexities

The EU's investment in the Circular Economy is best illustrated by the introduction of the European Green Deal (European Commission, 2020b; European Parliament, 2023b), which serves as a comprehensive framework aimed at making Europe climate-neutral by 2050 and ensuring a sustainable, resource-efficient economy (Fig. 3). This overarching policy integrates multiple strategies, regulations and directives, including those concerning waste shipments, plastics, batteries, and WEEE management. Within this framework, policies addressing waste management and CRM are designed to align with and support the Green Deal's broader objectives of reducing environmental impact and promoting resource efficiency.

In pursuit of this objective, the EU's Critical Raw Materials Action Plan emphasises the importance of recycling in addressing resource demand, although much of the action plan is focused on primary resources. The Critical Raw Materials Action Plan dovetails with the Circular Economy Action Plan through its recognition of the potential of waste as a resource (European Commission, 2020a). It acknowledges the lack of publicly available information on the quantities of raw materials in consumer products, as well as the actual amounts of waste that leave Europe (European Commission, 2020a). As part of the Critical Raw

[3] In the context of this study, "upcycling" refers to the practice of collecting or salvaging items or parts of items before they are disposed of or recycled, and repurposing them for a different use or, in some cases, reusing them for the same purpose but within a new item.

[4] "Wicked problems" are large, complex, and entangled problems that display knowledge uncertainties in relation to causes and impacts. It is therefore unlikely that conventional linear approaches to problem analysis and policy design will generate robust, effective, and broadly supported strategies for improvement.

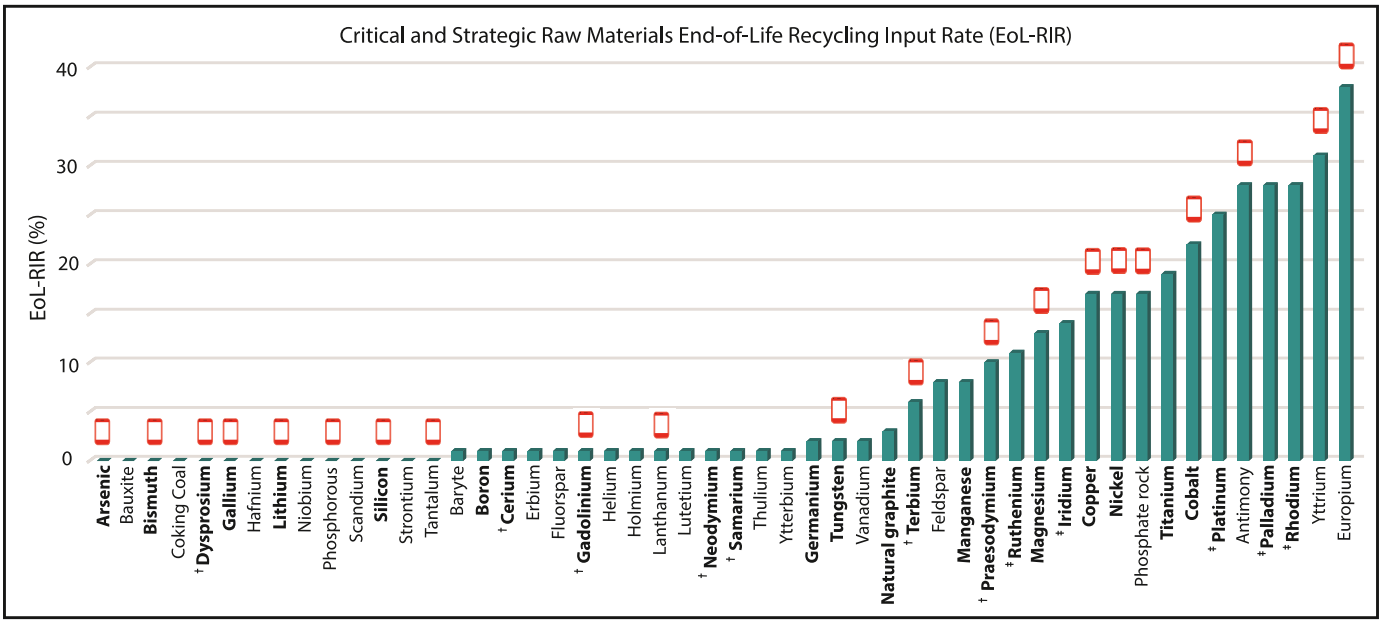

Fig. 2. Product end-of-life recycling input rate (EOL-RIR) for the 2023 CRM and SRM lists. SRM are bold. Data from (Talens Peiró et al., 2018). †REE; ‡PGE. CRM contained within cellular telephones are indicated by orange phone symbol.

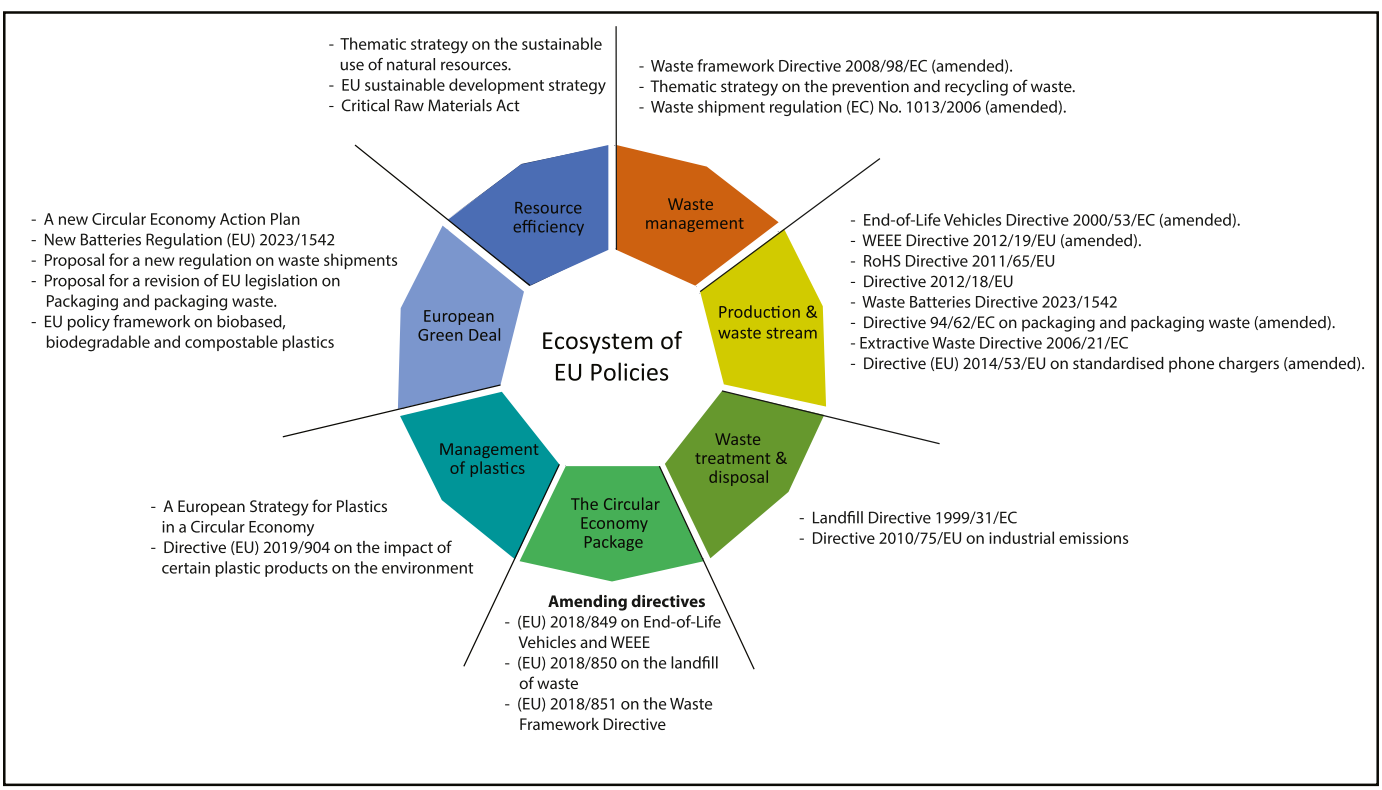

Fig. 3. The Green Deal ecosystem of European Union policies, strategies, and legislation surrounding waste, resource production, and the circular economy (Commission of the European Communities, 2005, Council of the European Union, 1999, European Commission, 2018b, European Commission, 2020b, European Commission, 2020c, European Commission, 2023a, European Parliament and Council, 2000, European Parliament and Council, 2008, European Parliament and Council, 2011, European Parliament and Council, 2012a, European Parliament and Council, 2023; European Commission, 2021; European Parliament and Council, 2012b; European Parliament and Council, 1994; European Parliament and Council, 2006b; European Parliament and Council, 2010; European Parliament and Council, 2018a; European Parliament and Council, 2018c; European Parliament and Council, 2019; European Commission, 2021b; European Commission, 2022a; European Commission, 2022b; Council of the European Union, 2009).

Materials Action Plan, the CRMA recognises that Europe has limited internal production of CRM, and is overly dependent on foreign CRM supply chains. It includes goals to ensure that the EU does not depend on a single non-EU country for more than 65% of its SRM, as well as to increase processing capacity to contribute at least 40% of the EU's SRM annual consumption by 2030 (European Commission, 2023a). The

CRMA also aims to boost recycling capacity to generate at least 25% of the EU's SRM annual consumption.

The EU has developed several strategies and associated directives recognising how WEEE utilisation could benefit the EU's supply of CRM and enhance the environment. These strategies, including those targeting plastics contained within WEEE, reinforce a more holistic approach to CRM retention, ensuring that all materials, not just metals, are efficiently managed and recycled. These strategies and directives (Fig. 3), serve as a framework for shaping production life-cycle policies that exert direct or indirect influence on the life cycle of CRM. Given the intricate nature of each branch within these legislative initiatives, along with their intended interoperability, this research specifically targets those most relevant to CRM and the EU's capacity to oversee, regulate, retain, and manage the production of both raw materials and WEEE.

4.1. Policies on WEEE collection and retention

The EU Waste Framework Directive 2008/98/EC (amended by Directive (EU) 2018/851 and Regulation (EU) 2023/1542) were developed in response to the Thematic Strategy on the Prevention and Recycling of Waste as a means of waste management (Commission of the European Communities, 2005). As a key piece of legislation in the EU's approach to waste management and resource efficiency, the EU Waste Framework Directive requires Member States to reduce waste generation and promote the recycling and recovery of raw materials from waste. It sets specific recycling targets for waste streams, including WEEE (European Parliament and Council, 2008).

EU Waste Framework Directive 2008/98/EC (European Parliament and Council, 2018b) is supported by WEEE Directive 2012/19/EU through the regulation of the *treatment*[5] and *collection*[6] of WEEE. It further introduced new amendments to increase collection targets of WEEE for all actors in the recycling process to 85% (replacing the previous target of at least 4 kg on average per inhabitant per year from private households or the same weight of WEEE collected in that Member State, on average, over the three preceding years, whichever is greater). Directive 2012/19/EU also contains provisions to address illegal WEEE exportation. Despite the legislative efforts, every member state failed to meet the 85% target for recycling by January 1, 2016, with less than 40% of all EU-generated WEEE collected and treated by that date (European Parliament, 2023a). Under the Circular Economy Action Plan, amendments were subsequently made to WEEE and Landfill Directives (2012/19/EU, 2000/53/EC and 1999/31/EC, respectively) setting a new target of 65% recycling of municipal waste by 2035, and a reduction of landfill in municipal waste to 10% by 2035.

Regardless of the compliance challenges surrounding Directive 2012/19/EU, most regulation has proven to be an effective driver of recycling, both on a large scale and for specific materials. For example, the EU End-of-Life Vehicle Directive 2000/53/EC (amended by Directive (EU) 2018/849) and the Commission Decision 2005/293/EC, places increased emphasis on the recovery of Fe, Al, and copper (Cu) from waste vehicles. However, enabling successful WEEE recycling depends largely on the consistent availability of WEEE with known concentrations of elements to process. The large uncertainty in the EOL-RIR creates challenges in monitoring the flow of waste inputs and outputs, since items, even large ones like combustion- or electric-engine vehicles, cannot be tracked if they do not enter the waste stream. Uncertainty in elemental content of recycling input detracts from resource estimation and recovery of mandated metals like Fe, Al, and Cu, as well as other CRM.

In addition, Pb, a relatively toxic metal, currently has a >75% recovery rate in the EU as it is regulated under the EU RoHS Directive 2011/65/EU (updated from the original RoHS Directive, 2011/65/EU) and, importantly, is associated with strict financial and legal penalties if not adhered (European Parliament and Council, 2011; Graedel et al., 2011). To help address the issue of tracking and to enable policy enforcement, the CRMA creates a framework to effectively label and track products that incorporate magnets and their SRM contents (European Commission, 2023a). However, magnets are only one component in WEEE devices.

4.2. Efficiencies and deficiencies of WEEE manufacturing

Whilst the Circular Economy Action Plan does provide a policy framework for recycling and recovery of materials from Li-ion batteries and construction materials (European Parliament, 2022), these strategies omit specific policies on improved material or CRM recovery from WEEE as a whole (European Commission, 2020c). The 2020 Circular Economy Action Plan acknowledges WEEE as one of the fastest-growing waste streams in the EU but varies in its recommendations for addressing it. Consumer-targeted initiatives are emphasised, including reducing planned obsolescence timelines, improving rights to repair, and promoting better recycling uptake by the public. However, it largely neglects the potential for smart manufacturing to enable more efficient recycling or improve CRM recovery from WEEE. The 2020 Circular Economy Action Plan does emphasise the pivotal role of design in mitigating environmental impact in various industries, including textiles, construction, plastics, and electronics and battery manufacturing. There remains, however, a visible absence of policy measures addressing the need for improved CRM retention.

Research into smart manufacturing measures have largely focused on waste reduction (see Blömeke et al., 2020; Abubakr et al., 2020; Ciliberto et al., 2021). Less research has focused on product and manufacturing design to engineer for efficiency in recyclability that is specific for material retention throughout the product lifetime. Component heterogeneity is recognised as a significant obstacle to recycling, as illustrated by the multitude of elements used in a common smartphone (Fig. 1). For example, wireless charging components in smartphones consist primarily of Cu coils and Nd magnets which are difficult to effectively separate. Material heterogeneity in printed circuit boards and inclusion of toxic or carcinogenic components in products (Birloaga and Veglió, 2018; Huang et al., 2009) further complicate effective recycling. For example, regulations including Waste Shipments (EC) No. 1013/2006, whilst progressive in their design to manage hazardous materials, can impede the efficient recycling of e-waste by creating logistical challenges and increasing costs for recyclers who need to transport waste to specialised facilities, often located in different countries (European Parliament and Council, 2006a). Additional research to develop more efficient recycling strategies and new methods of initial construction, would appear to be required.

4.3. Other barriers to secondary resource processing

Tracking waste streams has been identified as a policy challenge that obstructs the efficient recovery and treatment of valuable secondary metals as well as hazardous waste (European Commission, 2015; Cucchiella et al., 2015). Waste, particularly WEEE, is difficult to track because of EOL-RIR variability, technological developments, and illegal waste movement. The EOL-RIR is a complex function of recycling infrastructure, regulations, and market dynamics. Different regions may have different rates of recycling for specific materials, impacting the overall recycling input rate. WEEE evolves over time due to intrinsic changes in the items discarded into the waste stream, which themselves evolve as technology develops (Huisman et al., 2017). Tracking WEEE is

[5] In the context of this study, "treatment" refers to the recovery or disposal operations, including preparation prior to recovery or disposal – adopted from EC 2023/0079 (COD).

[6] In the context of this study, "collection" refers to the gathering of waste, including the preliminary sorting and preliminary storage of waste for the purposes of transport to a waste treatment facility – adopted from EC 2023/0079 (COD).

further complicated by the illegal movement of waste across borders, both within the EU and to non-EU countries.

The proliferation and widespread dispersion of small-scale, informal recycling activities exacerbate the environmental and social consequences (Zhang et al., 2015). According to the United Nations Environment Programme (2015), informal recycling activities are prevalent in many developing countries due to the absence of formal waste management systems at the national and local levels. These operations often lack rigorous regulation and safety measures, posing risks to human health and the environment. The report found that the waste streams fuelling these activities often originate from illegal practices such as WEEE trafficking. Such illicit practices have been estimated to account for approximately 60% of WEEE exported from the EU (Basel Action Network, 2018), despite prohibitions under the Waste Shipment Regulation (EC) No. 1013/2006.

5. Discussion

The intricate ecosystem of European resource policy (Fig. 3) is envisioned to balance the interconnected network of institutions, stakeholders, and processes that influence the production, consumption, and recycling of materials. However, the complexity of this materials ecosystem requires equally complex policy responses. Policy changes may result in feedback-loops, both positive and negative. Policy makers must become cognisant of the many facets of this ecosystem, including technical, economic, environmental, and social, as well as their respective interconnectivities. Improved consideration should be given to the entire life cycle of CRM, avoiding over-emphasis on initial primary production or end-of-life secondary resources.

Whilst the European Green Deal provides a structured approach to resource management and sustainability across various sectors, its success hinges on the effective implementation of the supporting directives and regulations. As such, challenges with implementation often arise within these underlying directives and regulations that feed into the Green Deal framework. For example, although EU directives prioritise waste reduction and reuse, the emphasis on recycling, particularly in the context of retaining minor but highly valuable CRM, remains limited. The mass-based approach under directives like the End-of-Life Vehicles Directive 2000/53/EC and the Recast WEEE Directive 2012/19/EU, prioritise bulk materials which are easier to process and quantify, rather than the 'more difficult to extract' CRM present in smaller quantities.

This issue is compounded by the growing supply of WEEE, which outpaces the capacity and effectiveness (both existing and expected) of current recycling systems, as highlighted by an estimated 64% of WEEE in the EU remaining unaccounted for each year (Eurostat, 2023). Complementary efforts to promote reuse, such as upcycling, are crucial but insufficient in isolation. The existing regulatory framework encourages component and material reuse but lacks standardised recycling efficiency requirements. As a result, inefficiencies in recycling processes lead to unnecessary CRM losses as exemplified by the high collection but low EOL-RIR/EOL-RR for many CRM as well the issue of non-functional recycling with regards to low-quantity CRM in recycled material (BIO by Deloitte, 2015; European Commission, 2023d). This is exponentially compounded by the lack of quantitative transparency relating to CRM in manufactured products (European Commission, 2020a). Simply put, without an understanding of how much goes "into" a product, one cannot truly know how much is supposed to "come back out".

Addressing these issues requires a shift in policy and regulatory frameworks to include more explicit targets for the recovery of CRM. We therefore propose the concept of forwardcycling (Fig. 4) to emplace mechanisms at the early stages of the product lifecycle for the purpose of CRM retention at the recycling stage. Forwardcycling involves design of electronic and electrical equipment utilising modular manufacturing techniques, to specifically facilitate the quantification, recycling and ultimately the retention of CRM. This can be implemented in three interdependent stages, namely holistic manufacturing, regulated labelling of materials used in produced parts, and integrated policies.

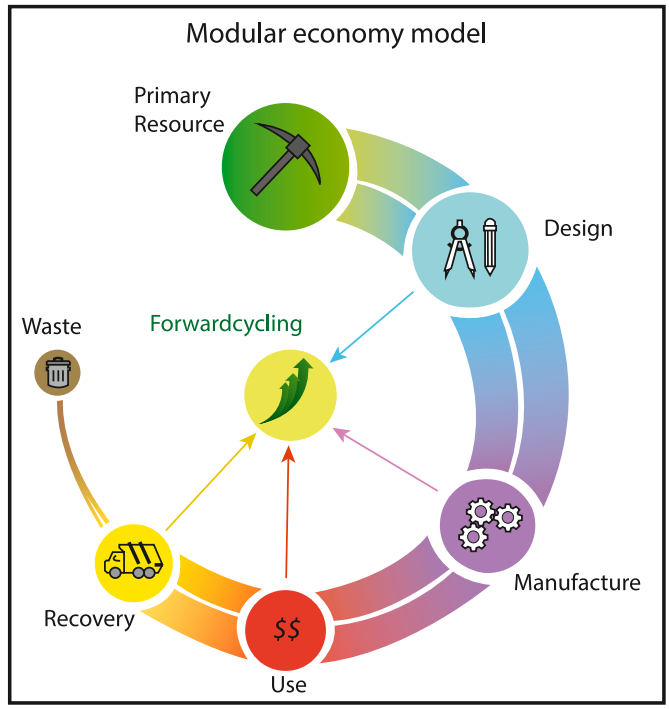

Fig. 4. A conceptual modular model for the circular economy integrating forwardcycling mechanisms. During the product design phase, accurate recording of critical raw material inputs is conducted for each manufactured module. This data is then traced throughout the manufacturing and usage phases. Upon reaching the end of the product life cycle, individual modules are retrieved and scanned to assess critical raw materials content before recycling, thereby determining an anticipated output value. Importantly, module tracking facilitates stringent exportation protocols, preventing the unauthorised export of critical raw materials-bearing WEEE.

5.1. Holistic manufacturing

To establish a resilient CRM supply chain, it is imperative to address the "by-design" loss of CRM inherent in current consumption and disposal practices. Primary and secondary resources should be viewed with similar economic potential, which can be achieved through the proactive integration of policies aimed at waste reduction and recycling efficiency into product design, specifically for material retention. This dovetails with the concept of modular production and design for disassembly and recycling (Fig. 4). This would allow simpler extraction and recycling of CRMs embedded within manufactured parts. Adoption of standardised fasteners and connections would further streamline the disassembly process, reducing time and recycling costs.

The concept of modular design in electronic devices is not novel and has been previously attempted by companies like Motorola Mobility LLC and Fairphone in efforts to reduce waste and harmonise interoperability between different cell phone companies through easily replaceable and upgradeable components (Proske et al., 2016; Sinha, 2021). However, policy mandates specifically geared towards modular design for the purpose of eventual recycling, ensuring that components have a quantified material value, are novel. Without such regulatory policies, innovative approaches like modular design struggle to compete with the profitability associated with the current economic model, which unwittingly promotes waste. Implementing forwardcycling at a European level could help establish standardisation, aligning with the EU's objectives outlined in the Proposed CRMA (European Commission, 2023a). A precedent for this has already been set by the "common phone charger" Directive 2014/53/EU (amended). This directive homogenised

phone chargers to a USB type-C cable that can be used irrespective of mobile phone brand. This policy directive successfully reduced waste associated with electronic device chargers by an estimated 11,000 tonnes per annum (European Commission, 2021b). Holistic manufacturing would effectively take Directive 2014/53/EU (amended) a step further, by mandating that chargers need to be manufactured in such a way as to make waste generated easily quantifiable and recyclable.

5.2. Regulated WEEE labelling

Providing quantitative data on CRM content for singular modular components, that can be listed and tracked throughout the various stages of the product life cycle, is a necessary component of holistic manufacturing. By implementing quantification and tracking parameters for each module, the unauthorised exportation of WEEE after use, could be more effectively measured and monitored. This would enable control of the exportation of WEEE modules and their contained CRM through regulated consent. Detailed information on CRM composition would also enable recycling facilities to accurately gauge process efficiency and evaluate the cost-effectiveness of various resource recovery methods. Clear labelling of the CRM content of materials used in products would significantly aid recyclers and help ensure that CRM are properly recovered and processed, minimising non-functional recycling. Through these measures, the EU could help ensure a continuous supply of CRM from recycled materials, reducing reliance on primary sources and enhancing resource security.

5.3. Integrated policies

Although modular design and standardised practices can significantly enhance CRM retention and recycling, these efforts need to be supported by properly integrated, comprehensive policy frameworks. Whilst the CRMA clearly articulates long-term quantified goals for CRM security of supply and provides a broad indication of what is needed to achieve these goals, it falls short in recognising the specific challenges to CRM retention and recycling. Significant focus is placed on the supply chain of CRM, particularly primary supply, whilst giving little emphasis to demand-side initiatives that encourage sustainable consumption practices. This was noted by Hoole et al. (2023) who pointed out that the CRMA does not specify where recycled materials originate, as they could be imported from external sources to meet production goals, thereby undermining a truly circular economy approach from within the EU. For example, the integration of the EU Emissions Trading System (EU ETS), which acts as a carbon pricing mechanism, with the Renewable Energy Directive (Official Journal of the European Union, 2023), has been effective in reducing emissions and promoting renewable energy. Similarly, by aligning the principles of modular design, with symbiotic policies such as the Waste Framework Directive with the Eco-design Regulation (Official Journal of the European Union, 2024), the EU could enhance CRM retention and recycling by penalising wasteful practices and incentivising the design of products that are easier to recycle.

As increased demand for most CRM is projected for the foreseeable future (see Carrara et al., 2023), the CRMA addresses supply challenges by focusing on finding more desirable mineral deposits and mine facilitation (e.g. Wall, 2014; Goodenough et al., 2018). However, quantifying CRM in WEEE, and forwardcycling policies for more effective product design, may be an even more effective means of securing the quantities of CRM required by the EU as well as offering other benefits, including increased strategic CRM autonomy and enhanced environmental protection.

6. Conclusion

The EU has taken a proactive stance in acknowledging its reliance on foreign countries for CRM, which are integral to its Green and Digital transitions, but existing policies have largely focused on waste reduction rather than CRM retention, leading to inefficiencies. This approach prioritises bulk materials over high-value CRM, breaking the circular economy model by neglecting the quality and recoverability of critical materials, especially in WEEE. By adopting a policy-backed approach that shifts toward forwardcycling and emphasises smart design, modularity, and enhanced material retention, the EU can significantly improve overall CRM recycling rates, thereby enabling more profitable economic industrialisation. The integration of this approach into existing frameworks like the CRMA, can strengthen the EU's strategic CRM autonomy, improve recycling efficiency, and support both economic and environmental goals.

CRediT authorship contribution statement

N.A. Vafeas: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization. **P. Slezak:** Writing – review & editing, Visualization, Resources, Investigation, Formal analysis, Data curation. **M.W. Hitzman:** Writing – review & editing, Validation, Funding acquisition, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known conflict of interests, competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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